

Lesson 5 — Design Challenge — Build-a-Bot Demo

Matches LEGO Teacher Portal lesson: "Island Explorers".

Objective (I can...): I can plan, build, and code a small machine that does a sequence I chose. **CS concept:** Applying sequence, decomposition, and pseudocode together · **Builds on:** Lessons 1-4. *Also serves as flex/catch-up — if earlier lessons ran long, use today to finish and consolidate.*

AI Connection (2 min): Building an AI feature is decompose → plan → build → test, the same loop you'll use today. *First-principles question:* "Which of those steps do you think AI engineers redo the most?"

Materials: LEGO CS & AI kit + device per pair, plan sheets, a simple challenge card. **Model already built** (Basics model from Lesson 2) — so the 30 minutes is plan + code + test, not build-from-scratch. Pairs may make small reconfigs only.

60-minute flow - Warm-up (8): Review carousel — fast prompts: What is a sequence? Decomposition? Pseudocode? - **Concept + demo (10):** Present the open challenge: make the model do 4+ ordered actions of your choice. Coach models pseudocode for one idea, then hands off. - **Build/Code (30):** Pairs decompose, write pseudocode, and code a 4+ step sequence on the already-built model (small reconfigs allowed). Coach coaches, doesn't solve. *Swap at 15 min.* - **Challenge (7):** Add one "surprise" action (light/sound/extra move) and update the plan to match. - **Share + pack-up (5): Build-a-Bot expo** — a gallery walk: bots on trays, pairs rotate to see each other's and each pair names the concept that helped most. **Coach photographs each build for the unit parent note.**

Pair roles: Co-plan, then Driver codes / Navigator checks plan; swap at 15 min. *(Trio rotates plan/code/test.)*

Differentiation: *Support* — a 3-action target plus a partial plan to finish. *Extension / fast-finisher* — require 6 actions with a light and a sound. *Reteach* — hand them their Lesson 4 pseudocode as a starting template.

Exit check: Pair delivers a working 4+ step program with matching pseudocode and can explain their plan. **Progress layer:** Send home the **Unit 1 parent note** ("your child can plan and code a sequence") with the **build photo**. **At-home (optional):** Sketch a dream robot and label 4 things it does in order. ~10 min.